

Optimizing Deucalion's Automatic Report Generation

High Performance Data Analysis 25/26

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- The Deucalion Team generates periodical reports:
 - Completed and failed jobs
 - Compute hours per partition
 - Jobs per partition
 - ...
- Current Apache Spark implementation is slow and must be optimized





- Standard iterative profiling-oriented optimization methodology

System Metrics



Runtime



CPU usage



Read and written bytes



Transmitted and received bytes

Apache Spark Metrics



Query plans



Jobs, stages, and tasks



Shuffle reads and writes

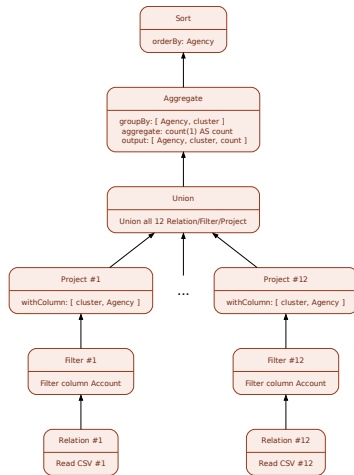
- 48-thread runs on single nodes of Deucalion's ARM partition (Fujitsu A64FX)
- 5 warmup runs, 5 measurement runs



Many query inefficiencies already optimized

BUT

Too many Spark actions (21)



Query plan for counting EuroHPC jobs per cluster



Single Query

```
SELECT Period, cluster, Agency, COMPLETED, COUNT(*), SUM(totalJobSeconds)
FROM jobs
GROUP BY Period, cluster, Agency, COMPLETED
```

Example Output Row

Period	cluster	Agency	COMPLETED	COUNT(*)	sum(totalJobSeconds)
Jan	ARM	EHPC	COMPLETED	10	3600



FastLoad

- Fixed predefined schema
- Bulk CSV loading



DuckDB

DuckDB

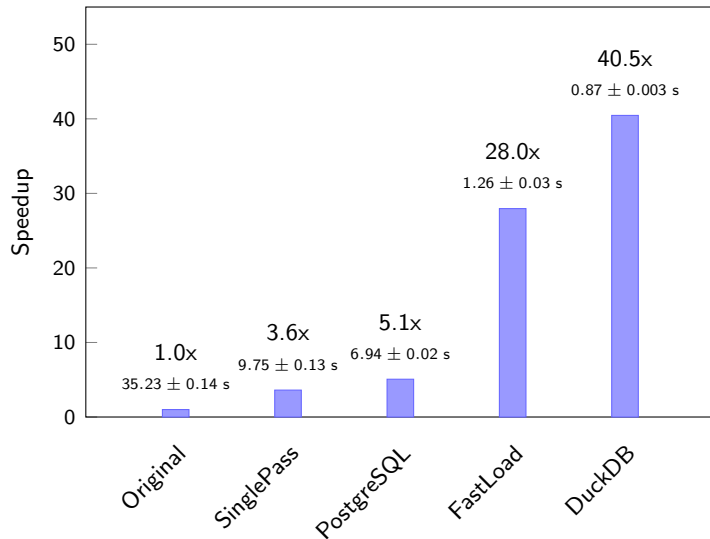
- Direct port of FastLoad



PostgreSQL

- No bulk CSV loading support
- Direct port of SinglePass

Results – Report Generator Performance



- Original → SinglePass:
83% reduction in disk reads

- SinglePass → FastLoad:

26 stages, 61 tasks



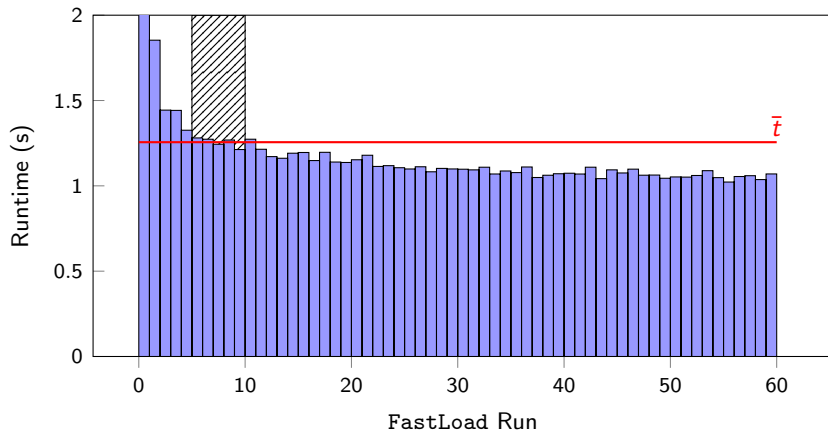
2 stages, 25 tasks

- DuckDB and PostgreSQL:
Different frameworks

Results – Methodological Problems



- Unrealistic methodology (no Spark init time, JVM warmup, page cache, ...)
- Cannot capture best-case Spark performance



Use the right tool for the job



SinglePass
3.6× speedup

No compromises



FastLoad
28.0× speedup

No schema inference



DuckDB
40.5× speedup

Platform change



PostgreSQL
5.1× speedup

ACID guarantees

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